CSS

Learning CSS
(Cascading Style Sheet)

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Cascading Style Sheet (CSS-3)

CSS describes how an HTML file should be displayed.

CSS is used to change the layout of an HTML page.

CSS consist of following:

1. Selector & Declaration Block
2. Property & Value

.....

H1 = {color:blue}

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Inserting inline CSS

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Document</title>
  </head>
  <body>
    <p style=“color:blue”>
      This is a dummy text and can be replaced.
    </p>
  </body>
</html>
```

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Inserting internal CSS

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Document</title>
    <style>
      p {
        color: blue;
      }
    </style>
  </head>
  <body>

    <p>Paragraph 1 - This is a dummy text and can be replaced</p>
    <p>Paragraph 2 - This is a dummy text and can be replaced</p>

  </body>
</html>
```

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Inserting external CSS

```
<html lang="en">
  <head>
    <link rel="stylesheet" href="styles.css">
  </head>
  <body>
    <p>This is a dummy text and can be replaced.</p>
  </body>
</html>
```

.....

styles.css

```
p {
    color: red;
  }
```

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Order of precedence

1. Inline style
2. Internal style
3. External style
4. Browser default

Note: If external stylesheet is referred after internal block then external style will take precedence

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Select based on tag, id or class

```
p {color: red;}
#p1{color: blue;}
.c1 {color: green;}
.c2 {color: yellow;}
p.c1 {color: maroon;}
```

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Refer more than one class

.c1 {color: green;}

.c2 {text-align: center;}

#p1{color: blue;}

p {color: red;}

.....

This is dummy text

.....

Id selector takes the first precedence and tag is last precedence

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Universal – affects every element

```
. * {
  text-align: center;
  color: blue;
}
```

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Grouping selectors

```
p, h1, div {
   color: red;
 }
```

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CSS Comments

```
/* This is a single-line comment */
H1 {
  color: blue;
}
```

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Colors and Backgrounds

```
div {
  background-color: lightblue;
 color:green
}
```

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Media queries

```
<style>
 .container {
   display: flex;
   flex-wrap: wrap;
 }
 @media only screen and (max-width: 600px) {
   /* For phones: */
   .box {
     color: green;
     width: 100%;
   }
 }
 @media only screen and (min-width: 600px) {
   /* For tablets: */
   .box {
     color: blue;
     width: 50%;
   }
 }
 @media only screen and (min-width: 768px) {
   /* For desktop: */
   .box {
     color: red;
     width: 25%;
   }
 }
 @media only screen and (orientation: landscape) {
   .box {
     background-color: lightblue;
   }
 }
 @media only screen and (orientation: portrait) {
   .box {
     background-color: lightgreen;
   }
 }
</style>
<body>
  <div class="container">
    <div class="box">Hello</div>
    <div class="box">Hello</div>
    <div class="box">Hello</div>
    <div class="box">Hello</div>
  </div>
</body>
```

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Borders and Margins

```
div {
    border: dashed 10px blue /*shorthand*/
  }
div {
    width: 100px;
    border: 5px blue solid;
    margin: 40px 40px 40px 40px; /* try auto and margin-top */
  }
```

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Text, Fonts

```
div {
    font-family: Arial;
    font-size: 14px;
}
```

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Position fixed

```
div {
    position: fixed;
    bottom: 0;
    right: 0;
    width: 500px;
    border: 10px solid blue;
  }
```

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Position sticky

```
div {
    position: sticky;
    top: 0;
    background-color: green;
    border: 2px solid #4caf50;
  }
```

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Padding

```
div {
      border:5px blue solid;
      padding: 10px 20px 30px 40px;
    }
```

.....

Try with just padding:10px and
Padding-top, padding-bottom, etc

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Background-image

```
body {
      background-image: url("colors.png");
      background-repeat: no-repeat; /* repeat  */
      background-position: center;   /* right top,left top */
      background-attachment: scroll;  /*fixed*/
      background-size:cover;
    }
```

.....

```
body {background: #eeeeee url("colors.png") no-repeat right top;}
```

.....

background-color, background-image, background-repeat, background-attachment, background-position

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Flex box

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      .container {
        display: flex;
        height: 100px;
        /*flex-direction: row;*/ /* try column */
        background-color: silver;
        /* flex-wrap: wrap; */
        /* flex-flow: row wrap; */
        /* justify-content: right; center left flex-start flex-end space-around space-between */
        /* align-items: center; flex-start flex-end stretch(default) */
        /* align-content: flex-start; */
      }

      .container > div {
        background-color: orange;
        width: 100px;
        margin: 10px;
        text-align: center;
        line-height: 75px;
        font-size: 20px;
      }
    </style>
  </head>
  <body>
    <div class="container">
      <div>HTML</div>
      <div>CSS</div>
      <div>JavaScript</div>
    </div>
  </body>
</html>
```

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Grid

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      .container {
        display: grid; /*inline-grid */
        background-color: silver;
        grid-template-columns: auto auto auto;
        column-gap: 20px;
        row-gap: 30px;
        /* gap: 20px 30px; for both row and col */
      }
      .container > div {
        background-color: orange;
        width: 100px;
        margin: 10px;
        text-align: center;
        line-height: 75px;
        font-size: 20px;
      }
    </style>
  </head>
  <body>
    <div class="container">
      <div>HTML</div>
      <div>CSS</div>
      <div>JavaScript</div>
      <div>HTML</div>
      <div>CSS</div>
      <div>JavaScript</div>
      <div>HTML</div>
      <div>CSS</div>
      <div>JavaScript</div>
    </div>
  </body>
</html>
```

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Styling Lists

```
<style>
    li {border: solid;}
    ul {
      border: solid;
      list-style-type: square; /* try circle upper-roman lower-alpha */
      /* list-style-image: url('colors.png'); */
      list-style-position: outside; /* try inside */
      margin: 20px;
      padding: 30px;}
  </style>
```

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Box model

```
div {
  width: 300px;
  border: 15px solid green;
  padding: 50px;
  margin: 20px;
}
```

Content > padding > border > margin

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Border Boxing

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      .c1 {
        width: 100px;
        border: 1px solid black;
        /* box-sizing: border-box; */
      }
      .c2 {
        width: 100px;
        padding: 10px; /*width becomes 120px */
        border: 1px solid orange;
        /* box-sizing: border-box; */
      }
    </style>
  </head>
  <body>
    <div class="c1">Hello World</div>
    <div class="c2">Hello World</div>
  </body>
</html>
```

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Pseudo class

```
a:link {color: green;}
    a:visited {color: red;}
    a:hover {color: orange;}
    a:active {color: blue;}
    div:hover {display: block; background-color: yellow;}
    input:focus {display: block; background-color: pink;}
    input:in-range {border: 2px solid red;}
```

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Form styling

```
input {
  width: 100%;
}

input[type=text] {
  width: 100%;
  padding: 12px 20px;
}

input[type=button], input[type=submit], input[type=reset] {
  background-color: #04AA6D;
  border: none;
  color: white;
}
```

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Wildcard Selectors (*, ^ and $)

The asterisk (*) matches any character, the caret (^) matches the start, and the dollar sign ($) matches the end of an attribute value.

.....

```
<style>
  [class*="t"] {
    background-color: yellow;
  }
  [class^="t"] {
    background-color: yellow;
  }
  [class$="t"] {
    background-color: yellow;
  }
</style>

<body>
  <div class="tree">This is tree class</div>
  <div class="stroke">This is stroke class</div>
  <div class="best">This is best class</div>
</body>
```

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Float

```
div{
    float:right /* left none */
  }
```

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Alignment

```
div {
    height: 200px;
    background-color: orange;
    color: white;
    text-align: center; /* left right justify */
     vertical-align: middle; /* top middle bottom */
   display:table-cell; /*needed for vertical align */
   }
```

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Vertical - Alignment

```
<div
style={{display:'flex',alignItems:'center',justifyContent:'center',backgroundColor:'yellow',height:'
100px'}}>

    <div >

      <button type="button" class="btn btn-primary">

        Centered button

      </button>

    </div>

  </div>
```

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Styling Tables

```
table,th,td {
      border: 1px solid; /* try border-bottom */
      border-collapse: collapse;
      text-align: center;
      vertical-align: middle; /*try bottom and top */
      padding:20px;
    }

tr:hover {background-color: orange;}

thtr:nth-child(even) {background-color: cyan;}

{width:100px; /* try percent also */}
```

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Gradients

```
div {

    /*background-image: linear-gradient(maroon, yellow);*/
    /*background-image: linear-gradient(to right,red,pink);*/
    /*background-image: linear-gradient(to bottom right,maroon, yellow);*/
    /*background-image: linear-gradient(90deg,maroon, yellow);*/
    /*background-image: linear-gradient(maroon, yellow,pink);*/
    /*background-image: radial-gradient(red, yellow, green);*/
    /*background-image: radial-gradient(maroon 10%, yellow 20%, green 30%);*/
    /*background-image: radial-gradient(circle,red, pink, green);*/
    background-image: repeating-radial-gradient(maroon, pink 10%, green 15%);

  }
```

Animation

```
<!DOCTYPE html>
<html>
<head>
<style>

@keyframes topics {
  from {top:0px;background-color:pink}
  to {top:200px;background-color:blue}
}
/*@keyframes topics {
  0% {background-color:pink}
  50% {background-color:yellow}
  100% {background-color:green}
}*/

div {
  width: 300px;
  height:200px;
  position:relative;
  margin:auto;
  animation-name:topics;
  animation-duration:5s;
  animation-delay:2s;
  animation-iteration-count:2; /* try infinite */
  animation-direction:alternate; /*reverse alternate-reverse*/
  animation-timing-function:ease-out; /*ease,linear,ease-in,ease-in-out*/
  animation-fill-mode: forwards; /*retains the style after */
  animation: topics 5s linear 2s infinite alternate;
}
</style>
</head>
<body>
<div></div>
</body>
</html>
```

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Transformation

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      .c {
        transform: translate(50px, 100px);
        /* transform:rotate(30deg) */
        /* transform: scale(1.2, 1.2); */
        /* transform: scaleX(1.3); */
        /* transform: scaleY(1.3); */
        /* transform: skew(10deg, 20deg); */
        /* transform: skewX(10deg); */
        /* transform: skewY(20deg); */
        /*transform: matrix(1, -0.3, 0, 1, 20, 40); */
        /* matrix(scaleX(), skewY(), skewX(), scaleY(), translateX(), translateY()) */
      }
    </style>
  </head>
  <body>
    <img src="flower.png" />
    <img src="flower.png" class="c" />
  </body>
</html>
```

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Dropdown Menu

```
ul {margin: 0px;padding: 0px;background-color: silver;}
li {display: inline-block;padding: 5px;}
li:hover {background-color: grey;}
.submenu {display: none;position: absolute;}
.menu:hover .submenu {display: block;}
```

.....

```
 <ul>
     <li>Home</li>
     <li class="menu">
       Courses
       <div class="submenu">
         <a href="#">Python</a><br>
         <a href="#">Java</a><br>
       </div>
     </li>
     <li>About</li>
     <li>Contact</li>
 </ul>
```

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Z-index

```
<style>
img {position: absolute;left: 0px;top: 0px;
 z-index: -1;}  /* try 1 2 3 just like stack */
</style>
<body>
<h1>This is a heading</h1>
<img src="meeting.jpg">
<p>This is a dummy text and can be replaced. This is a dummy text and can be
replaced.</p>
```

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Favi icon on title bar

```
<title>My Page Title</title>
 <link rel="icon" type="image/x-icon" href="/images/favicon.ico">
```

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CSS Colors

Color Names :
Tomato, Orange, DodgerBlue, MediumSeaGreen, Gray, SlateBlue
Violet, LightGray

HEX Value
#rrggbb, #ff6347, #000000, #ffffff

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Background-color - opacity

```
div {
  background-color: lightblue;
}
div {
  background-color: blue;
  opacity: 0.4;  // use below rgba to keep the text bold
}
div {
  background: rgba(0, 0, 128, 0.4)
}
```

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Overflow

```
div {
    width: 200px;
    height: 100px;
    background-color: yellow;
    overflow: auto; /* try hidden scroll visible */
  }

div {
    overflow-x: hidden;
    overflow-y: scroll;
  }
```

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Navigation Menu

```
ul{
   margin:0px;
   padding:0px;
   background-color:silver;
 }

 li{
   display:inline-block; /*try block for vertical menu */
   padding:5px;
   }

   li:hover{
     background-color:grey;
   }
```

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CSS Pseudo-elements (style parts)

```
div::first-line {color: blue;}
div::first-letter {color: red;font-size: 60px;}
p.lesson::first-letter {color: blue;font-size: 200%;}
p::before {content: url("colors.png");} /*try after */
::marker {color: blue;font-size: 23px;} /* for li items */
::selection {color: maroon;background: orange;}
```

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!important Rule

```
<style>
    #demo {color: blue;}
    .test {color: green;}
    h1 {color: red !important;}  /* try important in .test also */
  </style>

<body>
   <h1 id="demo" class="test" style="color: Orange">Hello World!</h1>
 </body>
```

.....

Inline styles, IDs, class, elements

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Math Functions

```
#div1 {
  /*width: calc(100% - 100px); */
  /*width: max(50%,100px); */
  width: min(50%,100px);
  border: 1px solid red;
  background-color: pink;
}
```

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Hero Image

```
.hero-image {
    background: url(officedesk.jpg) no-repeat center;
    background-size: cover;
    height: 500px;
    position: relative;}

.hero-text {
    text-align: center;
    position: absolute;
    top: 50%;
    left: 50%;
    transform: translate(-50%, -50%);
    color: white;}
```

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Shadow Effects

```
div {
    border: 1px solid blue;
    margin: auto;
    text-align: center;
    width: 400px;
    height: 200px;
    background-color: yellow;
    box-shadow: 2px 2px 4px red inset, 4px 4px 4px green;
    text-shadow: 2px 1px red;
  }
```

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Transitions

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background: maroon;
 /* transition: width 2s, height 2s, transform 2s; /* try without transition */
  /*transition-delay:1s;*/
  transition-property: width,height,transform;
  transition-duration: 2s,5s,1s;
  transition-timing-function:ease-out; /*ease-in,linear,ease-in-out*/
}
div:hover {
  width: 300px;
  height:200px;
  /*transform: rotate(180deg);*/
  /*transform:translate(100px,300px)*/
  transform:scale(2,2)
```

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Thank You

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